

Extended EDA of General Health and Demographic Variables in BRFSS 2

Date: 2026-04-23. This report focuses on the general health and fairness-oriented demographic variables that your team highlighted for the diabetes analysis: GenHlth, PhysHlth, MentHlth, DiffWalk, Age, Sex, Income, and Education.

The purpose of the report is to move beyond simple descriptive counts and show which variables exhibit clear gradients with diabetes status, which variables appear secondary, and which groups may deserve more attention in later prediction, fairness, or causal analyses.

The selected EDA dataset contains 253,680 respondents with no missing values across the variables used in this report. Diabetes prevalence is 13.9%, so the outcome is meaningfully imbalanced but still large enough for stable subgroup summaries.

The clearest health-status gradient appears in self-rated general health. Diabetes prevalence rises from 2.5% among respondents reporting excellent health to 37.9% among those reporting poor health, which makes GenHlth one of the most informative variables in this subset.

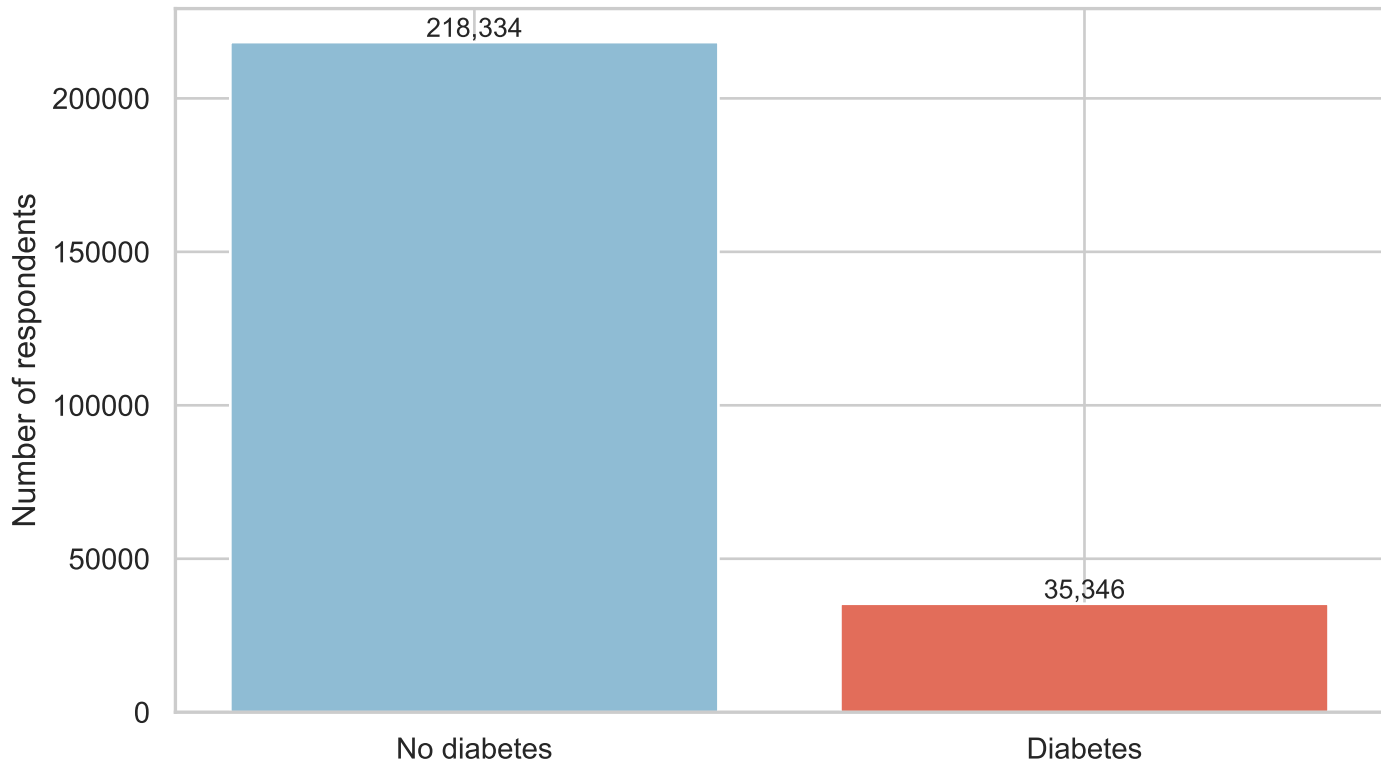
Dataset Overview

Metric	Value
Rows	253,680
Variables in report	9
Missing cells	0
No diabetes	218,334 (86.1%)
Diabetes	35,346 (13.9%)

Variable Dictionary

Variable	Type	Meaning
GenHlth	Ordinal	Self-rated health from excellent to poor
PhysHlth	Count	Poor physical health days in past 30 days
MentHlth	Count	Poor mental health days in past 30 days
DiffWalk	Binary	Serious difficulty walking or climbing stairs
Age	Ordinal	Age group 18-24 through 80+
Sex	Binary	Female/Male indicator
Income	Ordinal	Income category from <10k to 75k+
Education	Ordinal	Education from no school to college graduate
Diabetes_binary	Binary	No diabetes / diabetes

Outcome Distribution



High-Level EDA Interpretation

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Physical functioning variables are much more strongly separated by diabetes status than mental health alone. Difficulty walking increases from 13.5% in the non-diabetes group to 37.1% in the diabetes group, and mean poor physical health days rise from 3.64 to 7.95.

Mental health still shows a difference, but the magnitude is smaller: mean poor mental health days increase from 2.98 to 4.46, and the distribution remains heavily concentrated at zero in both groups.

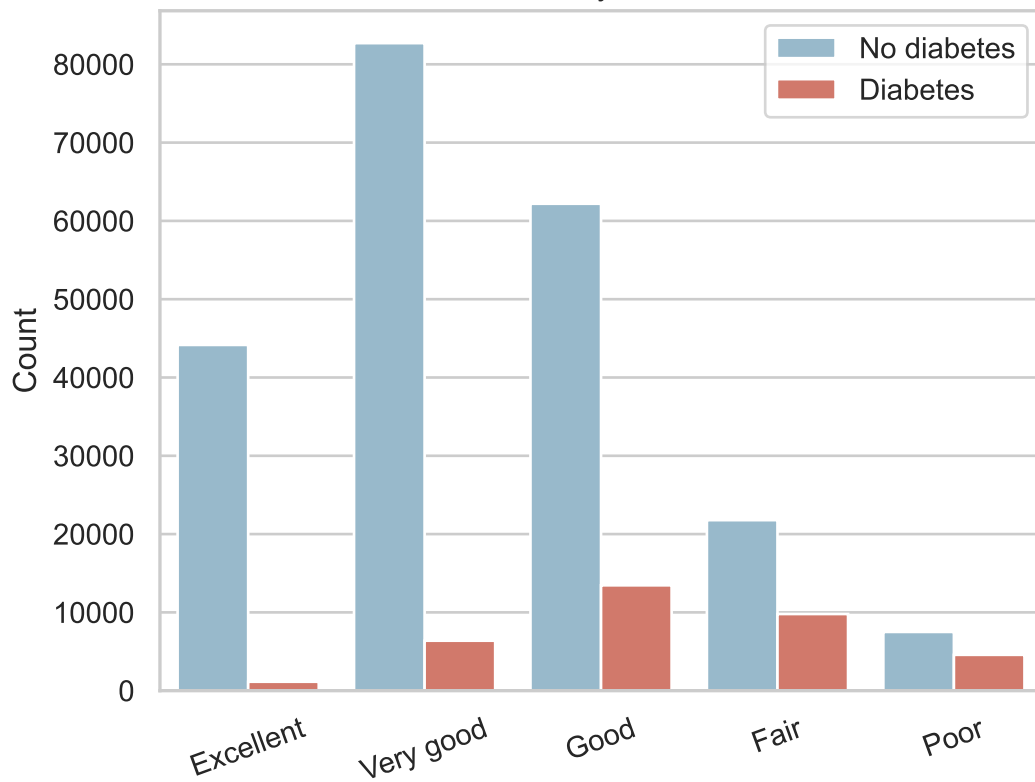
Demographic gradients are also pronounced. Diabetes prevalence increases across older age groups and decreases across higher income and education levels, which supports treating socioeconomic position as part of the main analytic story rather than as a minor adjustment variable.

Baseline Comparison by Diabetes Status

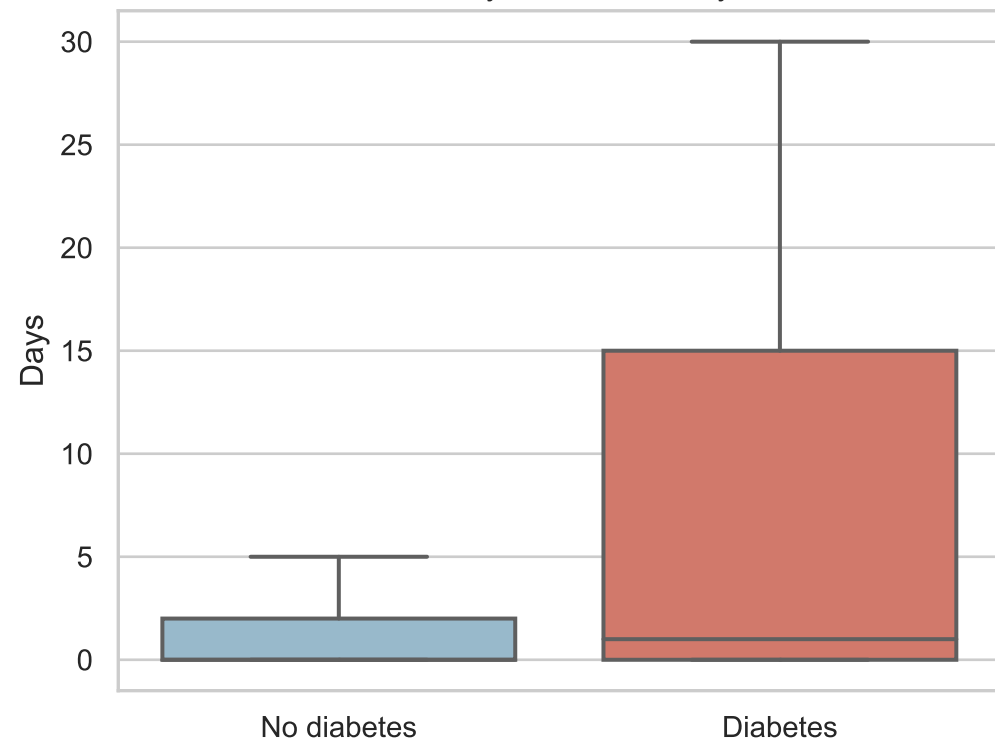
Variable	No diabetes	Diabetes	Difference	SMD
GenHlth	2.385	3.291	0.906	0.891
PhysHlth	3.641	7.954	4.313	0.439
MentHlth	2.978	4.462	1.484	0.184
DiffWalk	0.135	0.371	0.236	0.563
Age	7.814	9.379	1.565	0.571
Sex	0.434	0.479	0.045	0.091
Income	6.19	5.21	-0.98	-0.463
Education	5.1	4.746	-0.354	-0.35

SMD is included to show which variables separate the two outcome groups most strongly on a standardized scale.

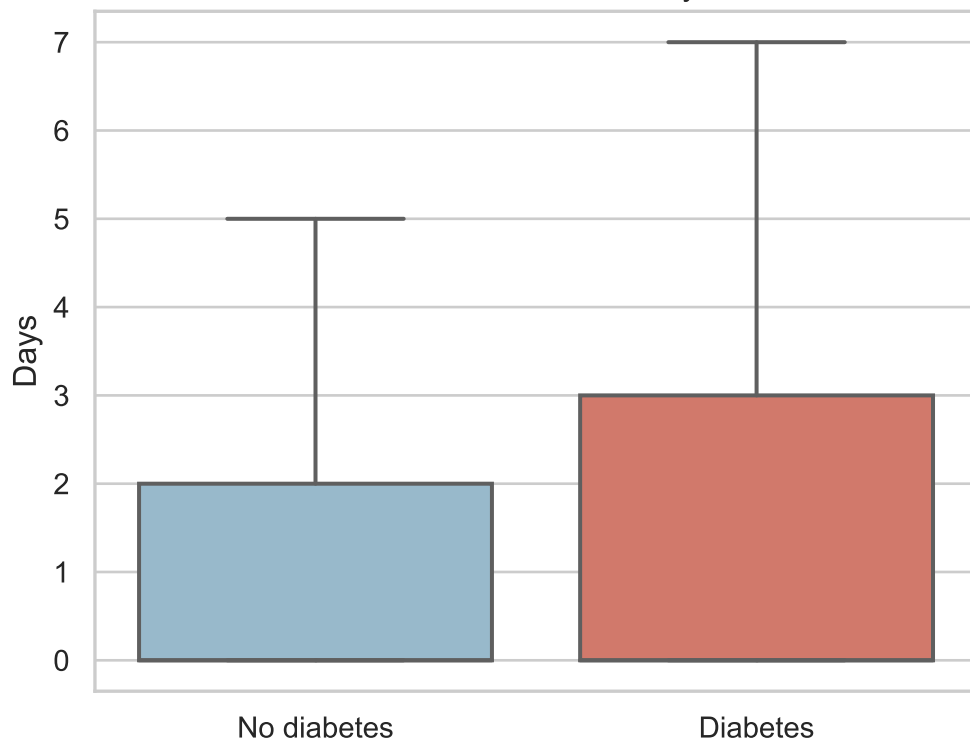
General Health by Diabetes Status



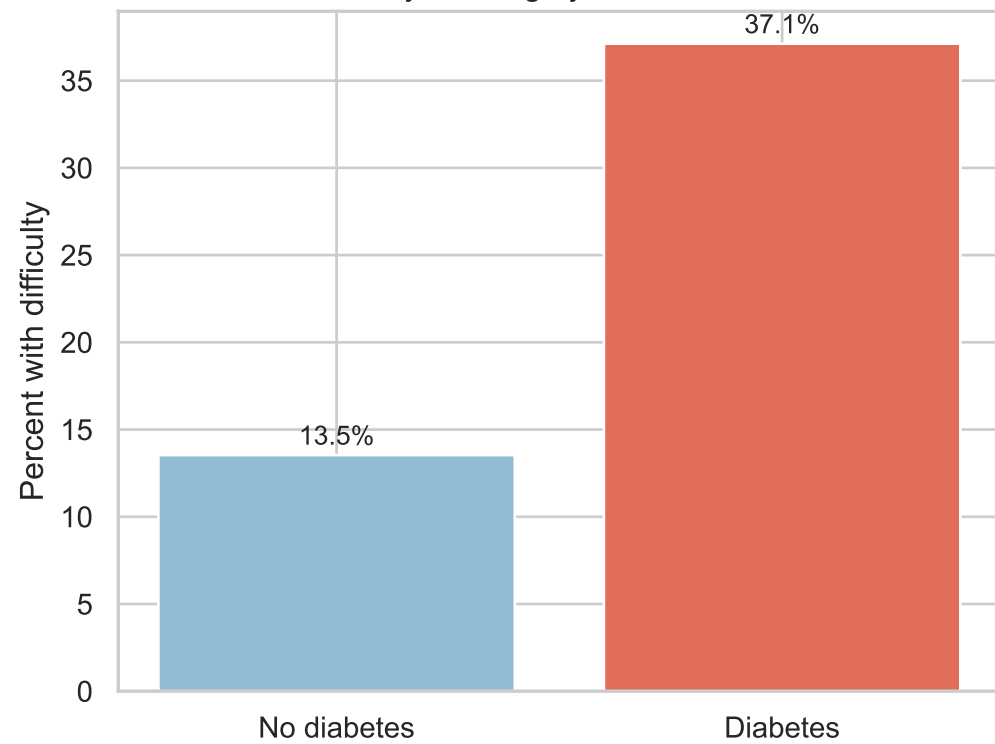
Poor Physical Health Days



Poor Mental Health Days



Difficulty Walking by Diabetes Status



Diabetes Prevalence by Self-Rated Health and Walking Difficulty

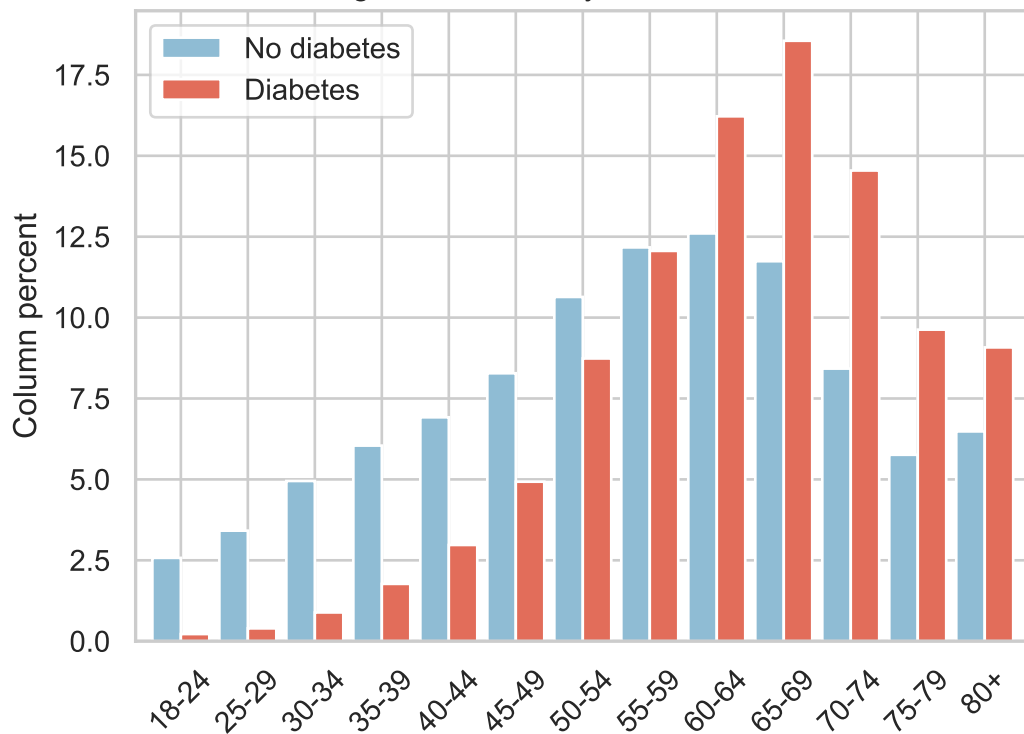
Category	Count	Diabetes prevalence	Variable
Excellent	45,299	2.5%	GenHlth
Very good	89,084	7.2%	GenHlth
Good	75,646	17.8%	GenHlth
Fair	31,570	31.0%	GenHlth
Poor	12,081	37.9%	GenHlth
No	211,005	10.5%	DiffWalk
Yes	42,675	30.7%	DiffWalk

Diabetes Prevalence by Poor Physical and Mental Health Day Buckets

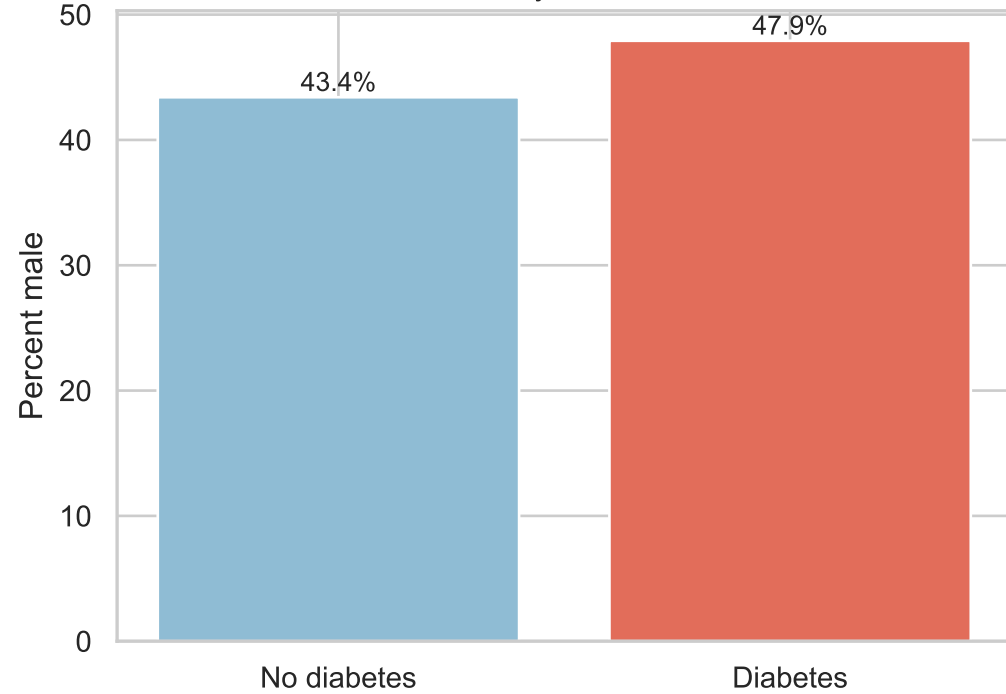
Category	Count	Diabetes prevalence	Variable
0	160,052	10.5%	PhysHlth
1-5	46,811	14.1%	PhysHlth
6-13	13,157	20.3%	PhysHlth
14-29	14,260	25.5%	PhysHlth
30	19,400	29.5%	PhysHlth
0	175,680	13.3%	MentHlth
1-5	41,792	11.9%	MentHlth
6-13	11,671	15.7%	MentHlth
14-29	12,449	19.2%	MentHlth
30	12,088	22.9%	MentHlth

Physical health days show a cleaner monotonic gradient than mental health days.

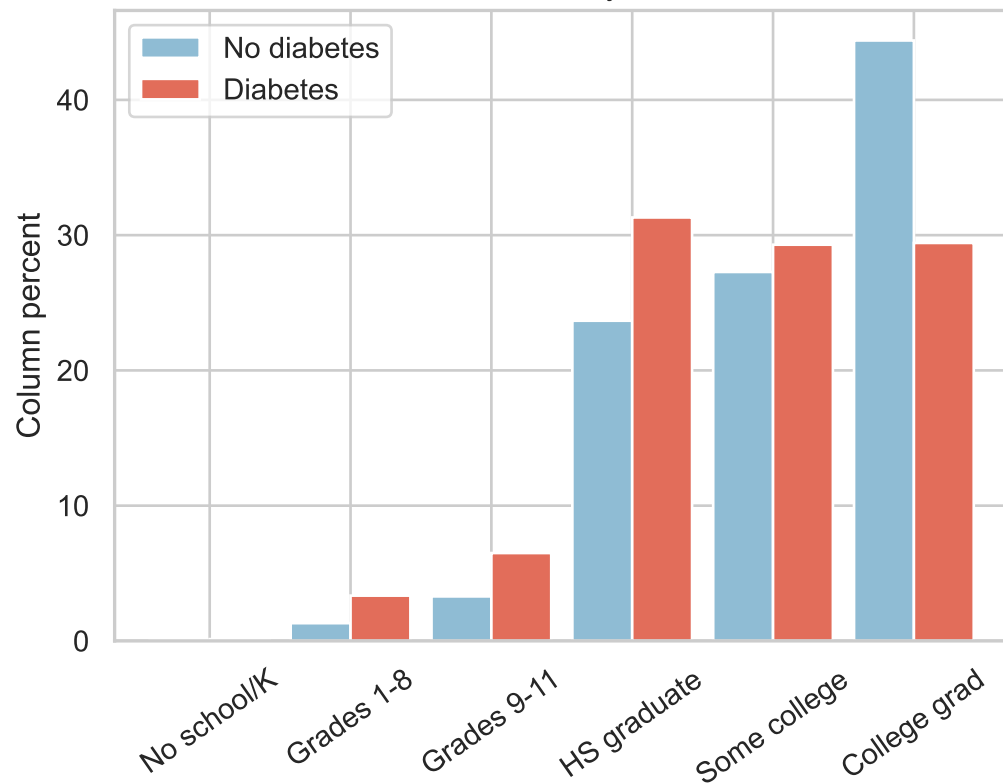
Age Distribution by Diabetes Status



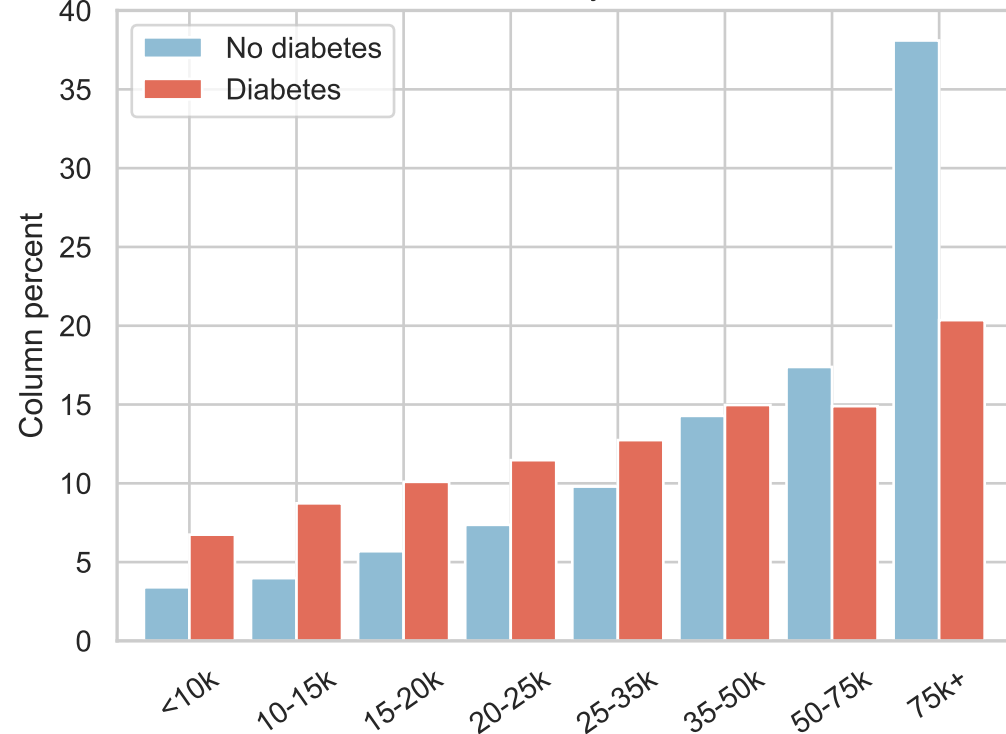
Male Share by Diabetes Status



Education Distribution by Diabetes Status



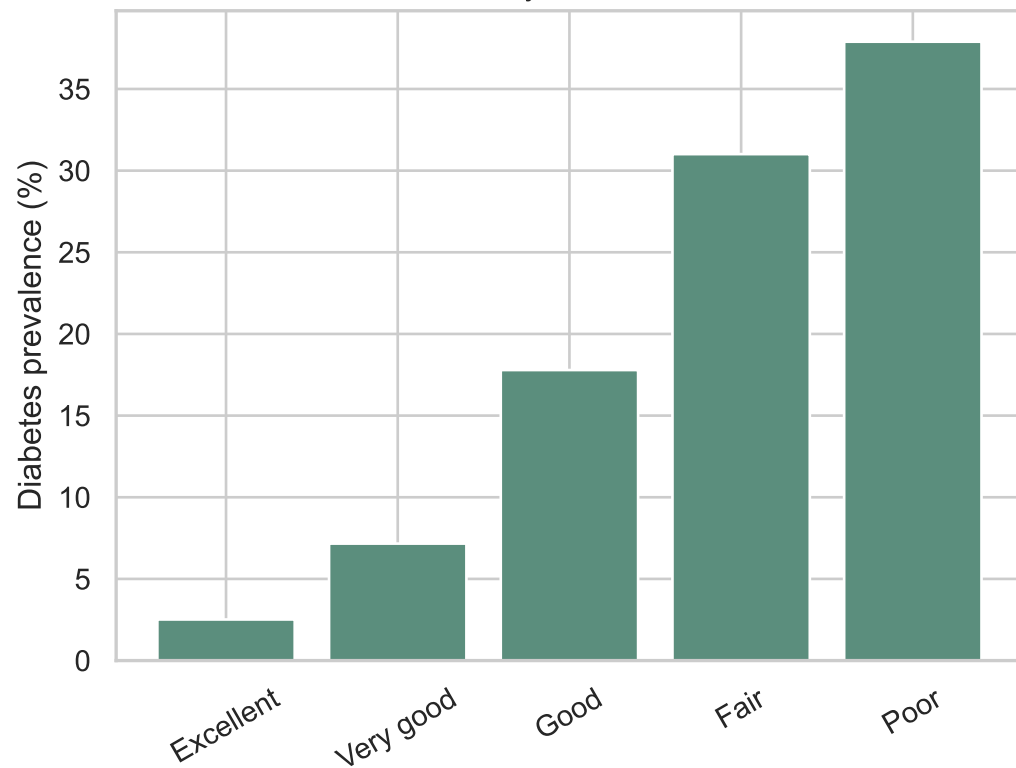
Income Distribution by Diabetes Status



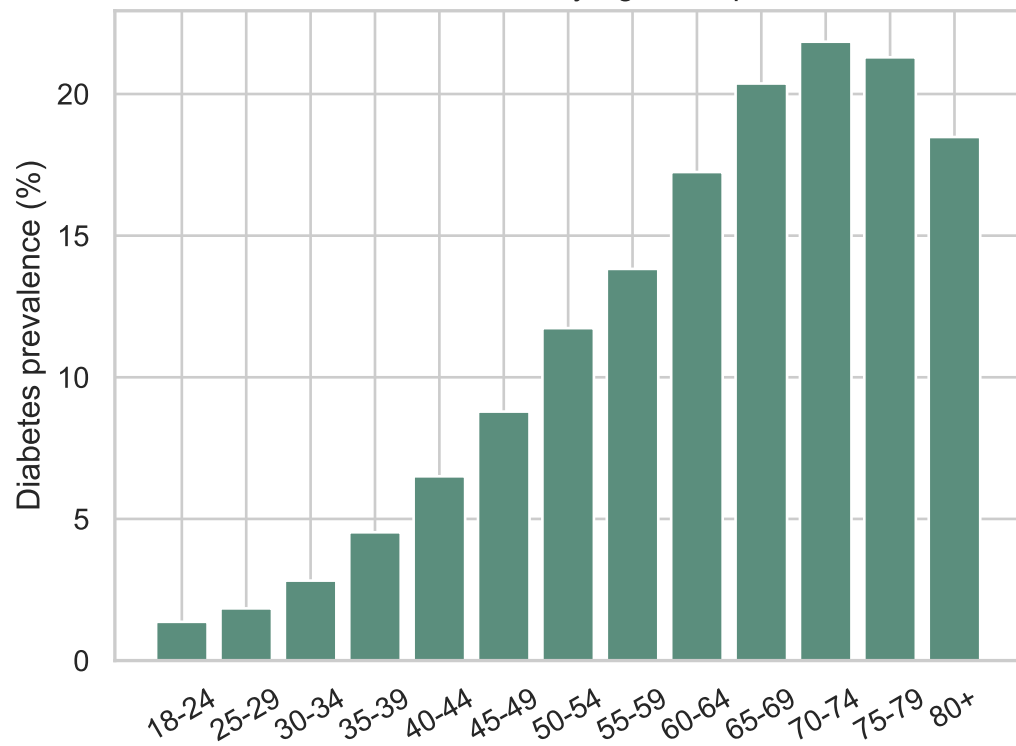
Diabetes Prevalence by Age, Sex, Income, and Education

Category	Count	Diabetes prevalence	Variable
18-24	5,700	1.4%	Age
25-29	7,598	1.8%	Age
30-34	11,123	2.8%	Age
35-39	13,823	4.5%	Age
40-44	16,157	6.5%	Age
45-49	19,819	8.8%	Age
50-54	26,314	11.7%	Age
55-59	30,832	13.8%	Age
60-64	33,244	17.2%	Age
65-69	32,194	20.4%	Age
70-74	23,533	21.8%	Age
75-79	15,980	21.3%	Age
80+	17,363	18.5%	Age
Female	141,974	13.0%	Sex
Male	111,706	15.2%	Sex
<10k	9,811	24.3%	Income
10-15k	11,783	26.2%	Income
15-20k	15,994	22.3%	Income
20-25k	20,135	20.1%	Income
25-35k	25,883	17.4%	Income
35-50k	36,470	14.5%	Income
50-75k	43,219	12.2%	Income
75k+	90,385	8.0%	Income
No school/K	174	27.0%	Education
Grades 1-8	4,043	29.3%	Education
Grades 9-11	9,478	24.2%	Education
HS graduate	62,750	17.6%	Education
Some college	69,910	14.8%	Education
College grad	107,325	9.7%	Education

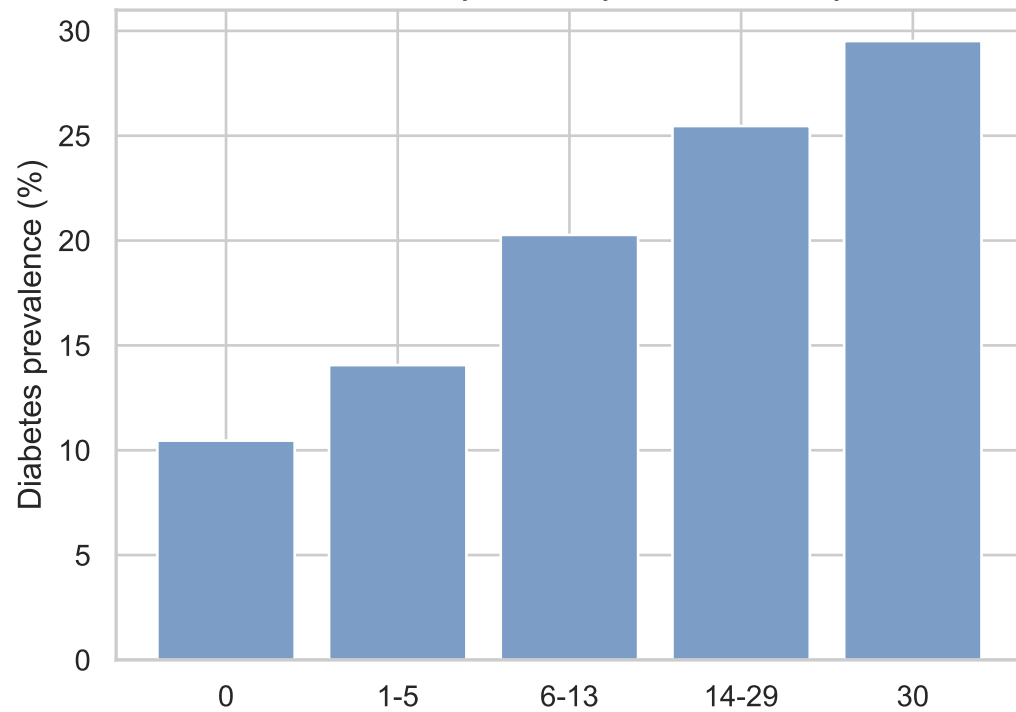
Prevalence by Self-Rated Health



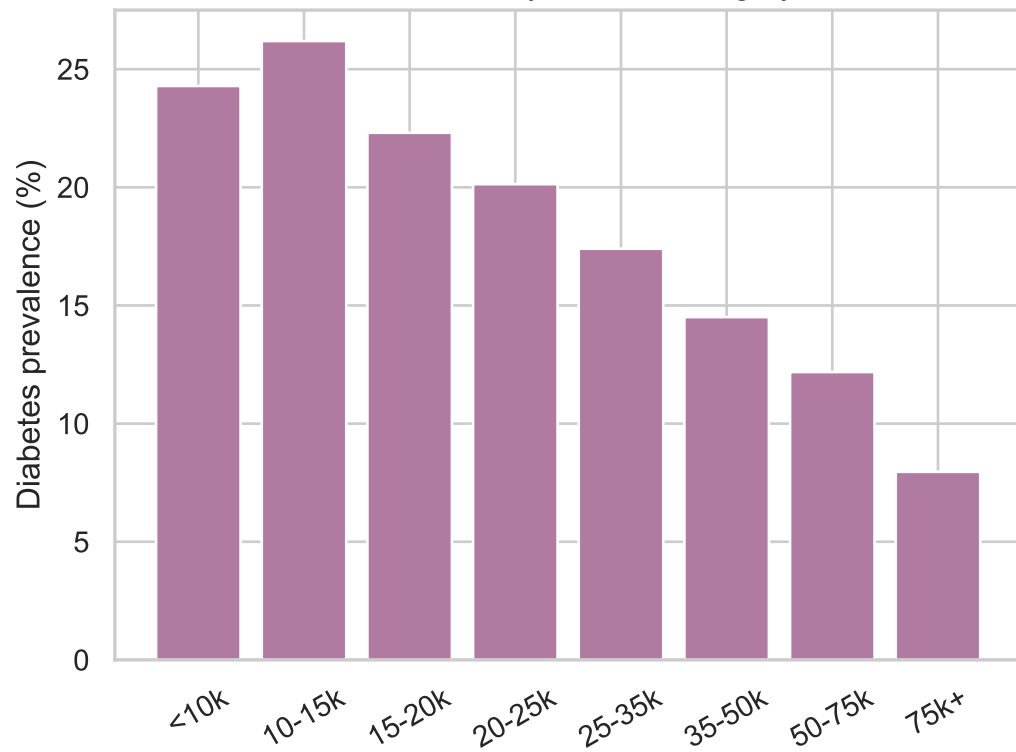
Prevalence by Age Group



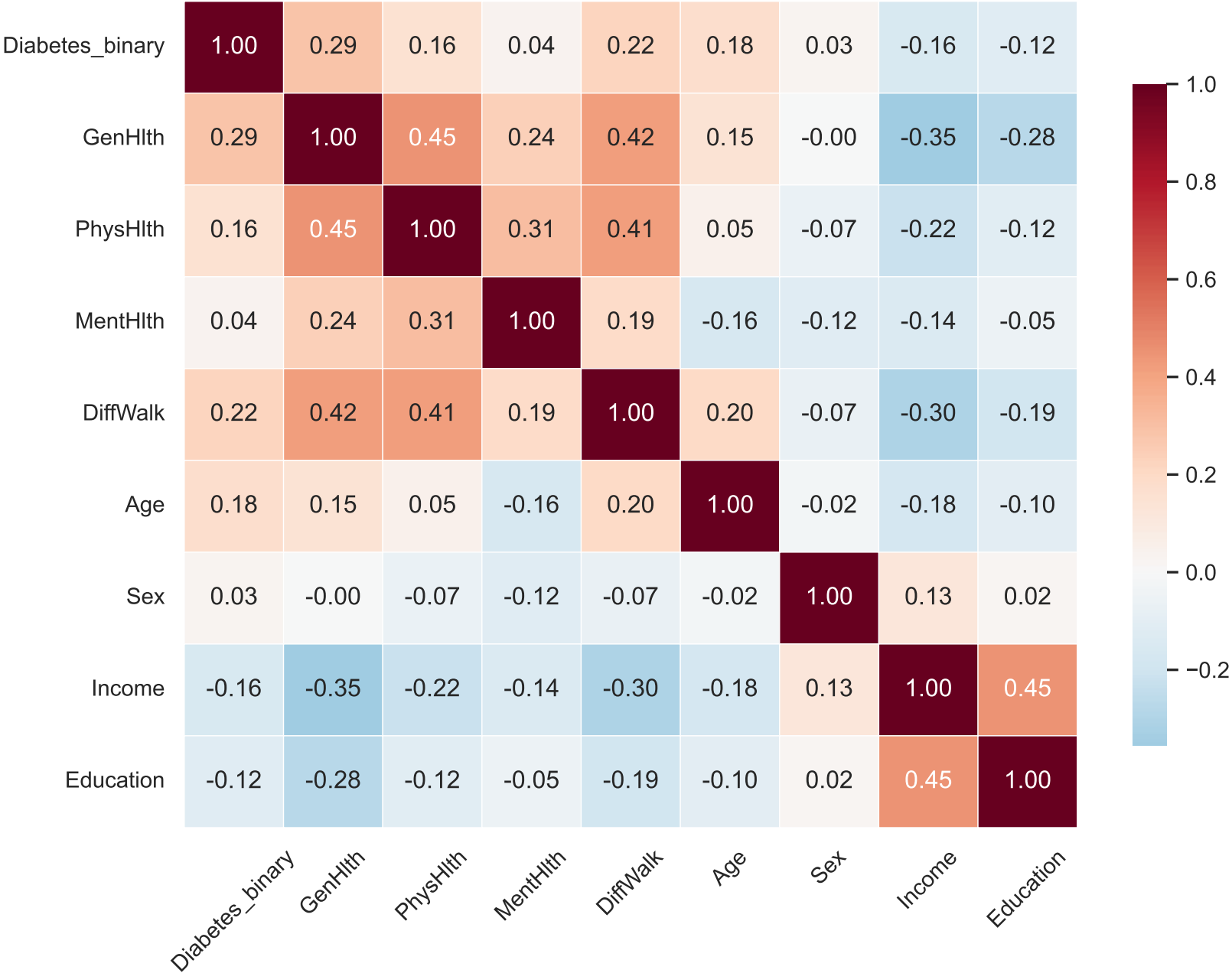
Prevalence by Poor Physical Health Days



Prevalence by Income Category



Spearman Correlation Matrix for Selected Variables



Spearman Correlation Matrix

Variable	Diabetes_bina	GenHlth	PhysHlth	MentHlth	DiffWalk	Age	Sex	Income	Education
Diabetes_bina	1.0	0.288	0.157	0.04	0.218	0.178	0.031	-0.163	-0.12
GenHlth	0.288	1.0	0.452	0.238	0.423	0.147	-0.004	-0.354	-0.276
PhysHlth	0.157	0.452	1.0	0.308	0.415	0.052	-0.068	-0.218	-0.115
MentHlth	0.04	0.238	0.308	1.0	0.188	-0.164	-0.12	-0.145	-0.054
DiffWalk	0.218	0.423	0.415	0.188	1.0	0.201	-0.07	-0.304	-0.188
Age	0.178	0.147	0.052	-0.164	0.201	1.0	-0.024	-0.176	-0.101
Sex	0.031	-0.004	-0.068	-0.12	-0.07	-0.024	1.0	0.127	0.024
Income	-0.163	-0.354	-0.218	-0.145	-0.304	-0.176	0.127	1.0	0.452
Education	-0.12	-0.276	-0.115	-0.054	-0.188	-0.101	0.024	0.452	1.0

Spearman correlation is more appropriate than Pearson here because the selected predictors mix binary, ordinal, and count scales.

Interpretation and Opportunities for Further Analysis

The main pattern in this subset is not a general mental-health story. Instead, diabetes appears most closely tied to a cluster of worse self-rated health, more poor physical-health days, and greater mobility limitation. Those variables are moderately correlated with one another, suggesting that they are capturing a shared health-burden dimension.

Income and education move in the opposite direction and show a clear socioeconomic gradient. That makes them useful not only as predictors but also as variables for subgroup performance checks, fairness discussion, and confounding adjustment if the team adds a focused causal analysis around physical activity later.

The correlation matrix belongs in the EDA section when it adds interpretation rather than clutter. In this case it helps because it shows both association with the outcome and redundancy among predictors, especially the link between GenHlth, PhysHlth, and DiffWalk, as well as the pairing of income and education.

Reasonable next steps include variable-block modeling, subgroup error analysis, and a small fairness section. A strong modeling comparison would be to test demographics alone versus demographics plus health burden versus demographics plus health burden plus mental health, and then quantify whether MentHlth adds meaningful predictive value beyond the stronger physical-health signals.